

SWYNEX – Network Security Analysis Report

Task: Task 2 – Network Security Analysis

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Tool: Wireshark 4.6.9

Environment: Controlled personal/lab network

1. Introduction

Wireshark was used to capture and analyze network traffic in a controlled environment. The analysis focused on identifying network protocols, observing unusual traffic patterns, and evaluating potential security concerns from a defensive perspective.

2. Objectives

- Identify network protocols in captured traffic.
- Analyze TCP, UDP and DNS traffic.
- Examine encrypted QUIC and TLS traffic.
- Identify TCP retransmissions and duplicate acknowledgements.
- Identify potential security concerns.
- Provide defensive recommendations.

3. Protocols Identified

Protocol	Observation
DNS	Domain-name resolution traffic was observed.
TCP	TCP communication, acknowledgements and analysis events were observed.
UDP	UDP traffic was observed.
QUIC	QUIC traffic with protected/encrypted payloads was observed.
TLS 1.2 / 1.3	Encrypted application data was observed over port 443.

4. Suspicious / Unusual Patterns

TCP Duplicate ACKs and TCP Fast Retransmissions were observed. These events can be associated with packet loss, congestion, delayed packets, or temporary network instability. They do not, by themselves, confirm a cyberattack.

5. Security Concerns

Encrypted traffic such as TLS and QUIC helps protect application confidentiality, but encryption can also make content inspection more difficult for security monitoring systems. Repeated retransmissions and duplicate acknowledgements may indicate network reliability issues and should be monitored.

6. Defensive Recommendations

- Monitor repeated retransmissions and duplicate ACKs.

- Use secure protocols such as TLS for protected communications.
- Keep network monitoring and analysis tools updated.
- Investigate unusual traffic patterns before treating them as malicious.
- Perform security testing only on systems for which authorization has been provided.

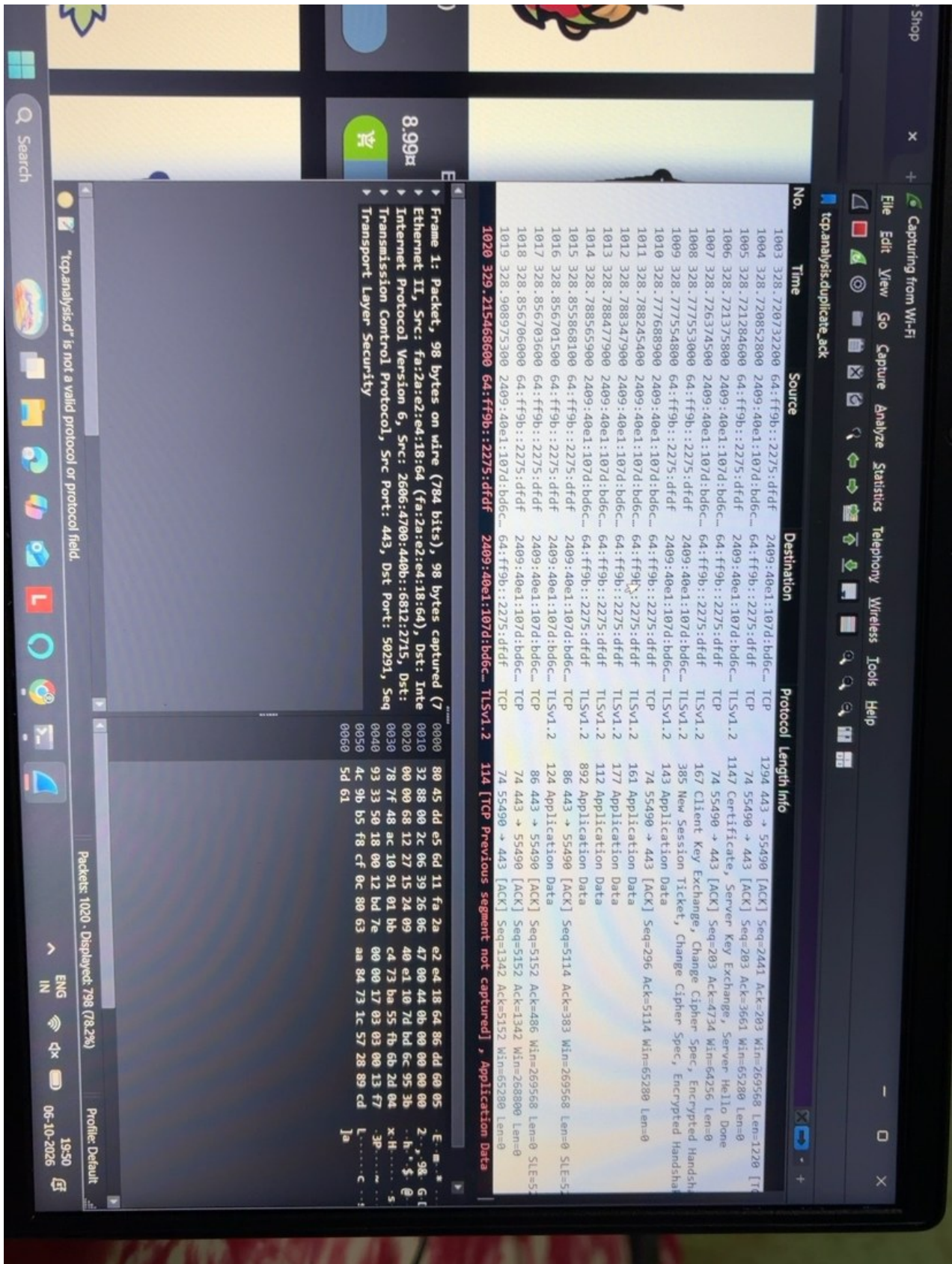
7. Conclusion

The captured traffic contained normal TCP, UDP, DNS, QUIC and TLS communications. TCP Duplicate ACKs and Fast Retransmissions were observed as potential network anomalies. No clear evidence of malicious activity was established from this limited capture. The exercise demonstrates how Wireshark can be used as a defensive tool to identify protocols and investigate potentially abnormal network behavior.

8. Evidence Screenshots

The following screenshots document the Wireshark observations supplied during the practical analysis.

TCP Duplicate ACK evidence



TCP Duplicate ACK evidence (additional capture)

Capturing from Wi-Fi

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tcp.analysis.duplicate_ack

No.	Time	Source IP	Destination IP	Protocol	Length	Info
266	59.84181970v	192.168.1.100	192.168.1.1	TCP	74	[TCP] Dup ACK 262#1] 50999 → 443 [ACK] Seq=3446 Ack=111
317	114.131821900	192.168.1.100	192.168.1.1	TCP	86	[TCP] Dup ACK 314#1] 54150 → 443 [ACK] Seq=3402 Ack=377
382	135.430976000	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 370#1] 443 → 54151 [ACK] Seq=5161 Ack=10
384	135.430977100	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 383#1] 443 → 54151 [ACK] Seq=5161 Ack=10
401	135.490888500	192.168.1.100	192.168.1.1	TCP	74	[TCP] Dup ACK 371#1] 443 → 54152 [ACK] Seq=5736 Ack=23
402	135.490888800	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 371#2] 443 → 54152 [ACK] Seq=5736 Ack=23
457	136.390191600	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 451#1] 54154 → 443 [ACK] Seq=2832 Ack=555
467	136.447703400	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 466#1] 443 → 54154 [ACK] Seq=5736 Ack=29
499	137.137224300	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 493#1] 54155 → 443 [ACK] Seq=3060 Ack=253
708	213.299422300	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 704#1] 443 → 61038 [ACK] Seq=1 Ack=1 Win=
725	214.043401500	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 712#1] 443 → 61038 [ACK] Seq=8973 Ack=253
729	214.043406000	192.168.1.100	192.168.1.1	TCP	66	[TCP] Dup ACK 726#1] 443 → 61038 [ACK] Seq=9138 Ack=276
757	225.938491000	192.168.1.100	192.168.1.1	TCP	74	[TCP] Dup ACK 754#1] 63165 → 443 [ACK] Seq=2791 Ack=101
995	328.666489400	192.168.1.100	192.168.1.1	TCP	86	[TCP] Dup ACK 989#1] 55489 → 443 [ACK] Seq=332 Ack=5363
1023	329.215540900	192.168.1.100	192.168.1.1	TCP	86	[TCP] Dup ACK 1019#1] 55490 → 443 [ACK] Seq=1342 Ack=5

Frame 266: Packet, 74 bytes on wire (592 bits), 74 bytes captured
Ethernet II, Src: Intel_e5:6d:11 (80:45:dd:e5:6d:11), Dst: fa:2a:e
Internet Protocol Version 6, Src: 2409:40e1:107d:bdc::953b:787f:48
Transmission Control Protocol, Src Port: 50999, Dst Port: 443, Seq

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0000  fa 2a e2 e4 18 64 80 45 dd e5 6d 11 86 dd 60 01  .a...d
0010  4a 6a 00 14 06 40 24 09 40 e1 10 7d bd 6c 95 3b  j...@
0020  78 7f 48 ac 10 91 26 20 01 19 00 fc 00 00 00 00  xH....
0030  00 00 00 00 00 02 c7 37 01 bb 8d c8 6b b9 fc f6  ....
0040  84 06 50 10 00 fb c2 1d 00 00                    ...P...

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QUIC traffic evidence

Quic Shop

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quic

No.	Time	Source	Destination	Protocol	Length	Info
198	46.815616300	2409:40e1:107d:bd6c...	2a03:2880:f368:120:...	QUIC	145	Handshake, DCID=aa003d3302491208
199	46.815866900	2409:40e1:107d:bd6c...	2a03:2880:f368:120:...	QUIC	96	Protected Payload (KPo), DCID=aa003d3302491208
200	46.816075200	2409:40e1:107d:bd6c...	2a03:2880:f368:120:...	QUIC	100	Protected Payload (KPo), DCID=aa003d3302491208
209	46.876021300	2a03:2880:f368:120:...	2409:40e1:107d:bd6c...	QUIC	101	Handshake, SCID=aa003d3302491208
210	46.876022200	2a03:2880:f368:120:...	2409:40e1:107d:bd6c...	QUIC	206	Protected Payload (KPo)
211	46.878136000	2a03:2880:f368:120:...	2409:40e1:107d:bd6c...	QUIC	334	Protected Payload (KPo)
212	46.878368900	2409:40e1:107d:bd6c...	2a03:2880:f368:120:...	QUIC	96	Protected Payload (KPo), DCID=aa003d3302491208
220	46.917437800	2a03:2880:f368:120:...	2409:40e1:107d:bd6c...	QUIC	110	Protected Payload (KPo)
243	48.204406900	2409:40e1:107d:bd6c...	2409:40e1:107d:bd6c...	QUIC	91	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
244	48.284289600	2001:4860:4827:7700...	2409:40e1:107d:bd6c...	QUIC	86	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
281	75.209375100	2409:40e1:107d:bd6c...	2001:4860:4827:7700...	QUIC	86	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
282	75.303181300	2001:4860:4827:7700...	2409:40e1:107d:bd6c...	QUIC	91	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
297	102.211168500	2409:40e1:107d:bd6c...	2001:4860:4827:7700...	QUIC	91	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
298	102.322763300	2409:40e1:107d:bd6c...	2001:4860:4827:7700...	QUIC	86	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
299	102.325561100	2001:4860:4827:7700...	2409:40e1:107d:bd6c...	QUIC	87	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
300	102.378660300	2001:4860:4827:7700...	2409:40e1:107d:bd6c...	QUIC	91	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
304	109.325782900	2409:40e1:107d:bd6c...	2001:4860:4827:7700...	QUIC	87	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
307	109.391542000	2001:4860:4827:7700...	2409:40e1:107d:bd6c...	QUIC	91	Protected Payload (KPo), DCID=fbff8b6dbc20d1a
453	136.326556800	2409:40e1:107d:bd6c...	2001:4860:4827:7700...	QUIC	91	Protected Payload (KPo), DCID=fbff8b6dbc20d1a

Frame 244: Packet, 86 bytes on wire (688 bits), 86 bytes captured
Ethernet II, Src: fa:2a:e2:ed:18:64 (fa:2a:e2:ed:18:64), Dst: Inte
Internet Protocol Version 6, Src: 2001:4860:4827:7700::, Dst: 2409
User Datagram Protocol, Src Port: 443, Dst Port: 55801
QUIC IETF

8.99M

Profile Default

TLS 1.2 / 1.3 evidence

